Jimena Medina Rubio

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EDUCATION

Institute of Marine and Atmospheric Research at Utrecht University Utrecht, Netherlands
PhD in Physical Oceanography August 2024 — August 2028
Studying the spatial distribution and temporal evolution of buoyant macroplastics on the ocean surface in the North Sea, driven by surface circulation, mesoscale and submesoscale dynamics, and air–sea interactions. The work integrates machine learning methods, observational data from surface drifters, and numerical simulations.

Imperial College London London, UK
MSci Physics with a Year Abroad (First Class Honours) October 2020 — June 2024
Solid foundation in physics and numerical methods, with practical experience gained through laboratory work and computational assignments. Relevant final-year elective courses: Hydrodynamics, Atmospheric Physics, Space Physics & Plasma, and Vortex Dynamics. MSci thesis on the simulation and analysis of transport pathways around Curaçao to determine the impact of pollutants on coral reefs, utilising methods from network theory (Grade: First Class).

Utrecht University Utrecht, Netherlands
Year 3 Exchange September 2022 — June 2023
Elective courses from the Climate Physics Master's: Geophysical Fluid Dynamics, Atmospheric Composition & Chemical Processes, Dynamical Meteorology, Dynamical Oceanography.

Ramiro de Maeztu High School Madrid, Spain
IB & Spanish Baccalaureate September 2019 — June 2020
43/45 final grade in the International Baccalaureate, 9.88/10 average in the Spanish Baccalaureate with honours mention. STEM–focused subjects including Mathematics, Physics and Chemistry

PUBLICATIONS

Jimena Medina-Rubio et al. "Using surface drifters to characterise near-surface ocean dynamics in the southern North Sea: a data-driven approach". English. In: *Ocean Science* 22.1 (Jan. 2026), pp. 49–74. ISSN: 1812-0784. DOI: 10.5194/os-22-49-2026

PRESENTATIONS

- *Buoyant Particle Transport and Accumulation in the North Sea: a Data-Driven Approach*; Ocean Sciences Meeting; Expected February 2026; Glasgow, Scotland [Poster]

- *Machine learning and surface drifters for analysing near-surface ocean dynamics in the southern North Sea*; BBOS Conference; October 2025; Antwerp, Belgium [Oral Presentation] Awarded 2nd Prize for Best Presentation.

- *Using surface drifters to characterise near-surface ocean dynamics in the southern North Sea*, CoPart-CoFlow summer school; June 2025; La Londe les Maures, France [Poster]

TEACHING EXPERIENCE

Utrecht University, Department of Physics
Teaching Assistant 2024 - 2026
- Turbulence in fluids (Y3 BSc course): led tutorials on the description of flow instabilities and their transition to turbulence using dynamical systems theory and statistical physics (25 students, 7.5 CTS).
- Dynamical Oceanography (MSc course): led tutorials on the explanation of the physical processes responsible for the ocean and shelf sea circulation (50 students, 7.5 CTS).

Master's thesis co-supervisor 2025 - 2026
Co-supervising MSc thesis (45 CTS) from a student of dual Climate Physics & Data Science Master's on the determination of eddy coherence from particle trajectories using ML methods.

WORK EXPERIENCE

Deltares Delft, Netherlands
Summer Research Intern July 2023 – September 2023
Participated in EDITO EU project, developing the D-EcoImpact tool, embedded within the DELFT3D hydrodynamic model, to draw maps of animal habitat suitability based on physical and biochemical data.

Imperial College UROP London, UK
Undergraduate Research Placement funded by NERC June 2022 - August 2022
Developed an ice-sheet fracture model in COMSOL to simulate the marine ice-cliff instability in Antarctica using the phase field method to locate damage propagation.

GRANTS, AWARDS & ACHIEVEMENTS

Second Prize for Best Presentation (BBOS) Antwerp (Belgium), 2025
Recognised by the jury for outstanding presentation quality and scientific contribution in the Oceans and Climate session at the BBOS conference on Climate Dynamics and Earth System Processes.

Organiser of Early Career Scientists Symposium Utrecht (Netherlands), 2025
Part of the organising committee of the *UU-NIOZ Early Career Scientist Symposium*, a two-day conference with 50 participants in the field of ocean sciences.

Recipient of Alan Turing Scheme (UK government) London (UK), 2023
UK government-funded global mobility programme supporting international study through university exchange placements (£3,700 stipend).

Recipient of NERC grant for Summer Research Placement London (UK), 2022
Research grant for 8 weeks (£415/week stipend) to work at the Grantham Institute at the Civil and Environmental Engineering Department.

First Prize in National STEM Pre-University Innovation Challenge Madrid (Spain), 2020
National innovation challenge organised by La Caixa Foundation to promote interdisciplinary STEM research for high school students. *Press coverage: El Mundo, June 2020*

International Olympiads of Metropolises Moscow (Russia), 2019
Selected as one of two students to represent Spain in the Physics delegation at the pre-university Olympiads.

SKILLS

- **Programming:** Scikit-learn, Parcels, Python, GitHub, DELFT3D, COMSOL.
- **Languages:** Spanish (Native), English (Fluent), French (Advanced), German (Basic).